

SPANNING TREES OF $G \times K_2$ BY VERTICAL SUPPORT, PARTIAL DOUBLES, AND THE BIJECTIVE PROGRAM FOR THE HYPERCUBE TREE FORMULA

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ABSTRACT. For a connected graph G , we count the spanning trees of the prism $G \times K_2$ by their *vertical support* — the set S of vertices of G whose vertical edge is used: the count is $\frac{1}{2} c(G) 2^{|S|} F_G(S)$, where $c(G)$ is the number of spanning trees and $F_G(S)$ the number of S -rooted spanning forests of G . Iterating this identity over the n directions of the hypercube $Q_n = Q_{n-1} \times K_2$ telescopes, through Pascal’s rule, to Stanley’s product formula $c(Q_n) = 2^{2^n - n - 1} \prod_k k^{\binom{n}{k}}$, giving an elementary inductive derivation of that formula and, more importantly, reducing the problem of a fully bijective proof — open since [2, Example 5.6.10], and still open in bijective form after Bernardi’s combinatorial proofs [1] — to a single per-level bijective step. We prove the full-support slice of that step by an explicit bijection, reformulate the general slice as a spanning-tree identity for *partial doubles* of G , report an exactly verified spin-uniformity phenomenon (Bernardi’s independence holds exactly for rooted-anywhere trees and provably fails for root-fixed ones), and record the experimental fine structure of the remaining step at the level of level-projections, including where its naive product law breaks.

1. BACKGROUND

Stanley’s *Enumerative Combinatorics* vol. 2 derives

$$(1) \quad c(Q_n) = 2^{2^n - n - 1} \prod_{k=1}^n k^{\binom{n}{k}}$$

from the Laplacian eigenvalues of Q_n and remarks (1st ed., p. 62, closing Example 5.6.10) that “a direct combinatorial proof of this formula is not known.” Bernardi [1] gave two combinatorial proofs — one via an independence property of edge-direction “spins” in uniformly random rooted forests, one via killing involutions — and the second edition of [2] (2024, Notes to Ch. 5) updates the remark: a *bijective* proof is not known. This paper organizes a route to one.

Throughout, G is a connected graph on M vertices (for (1), $G = Q_{n-1}$ and $M = 2^{n-1}$), and $G \times K_2$ is the prism over G : two copies $G \times \{0\}$, $G \times \{1\}$ (*horizontal* edges) plus a *vertical* edge $\{(u, 0), (u, 1)\}$ for each $u \in V(G)$. For a spanning tree T of $G \times K_2$, let $S(T) \subseteq V(G)$ be its vertical support. For $S \subseteq V(G)$, let $F_G(S)$ denote the number of spanning forests of G with exactly $|S|$ components, each containing one vertex of S ; by the all-minors Matrix-Tree theorem $F_G(S) = \det L_G[\bar{S}]$, the minor of the Laplacian with the rows and columns of S removed.

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2. THE SUPPORT-PRODUCT THEOREM

Theorem 2.1. *For every nonempty $S \subseteq V(G)$, the number of spanning trees T of $G \times K_2$ with $S(T) = S$ is*

$$N_G(S) = \frac{1}{2} c(G) 2^{|S|} F_G(S).$$

Proof. Weight the vertical edge at u by an indeterminate z_u and every horizontal edge by 1, and let $t(z) = \sum_T \prod_{u \in S(T)} z_u$ be the weighted spanning-tree sum. With vertices ordered level by level, the weighted Laplacian is $\begin{pmatrix} L_G + Z & -Z \\ -Z & L_G + Z \end{pmatrix}$, $Z = \text{diag}(z_u)$, which the orthogonal level-swap basis $\frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}$ block-diagonalizes as $L_G \oplus (L_G + 2Z)$. For a weighted graph on N vertices the product of the nonzero Laplacian eigenvalues is N times the tree sum, and the kernel lies in the L_G block; with $N = 2M$,

$$t(z) = \frac{1}{2M} \left(\prod_{\lambda \neq 0} \lambda(L_G) \right) \det(L_G + 2Z) = \frac{c(G)}{2} \det(L_G + 2Z).$$

Expanding by principal minors in the diagonal matrix $2Z$ and applying the all-minors Matrix-Tree theorem, $\det(L_G + 2Z) = \sum_R 2^{|R|} F_G(R) \prod_{u \in R} z_u$. Both sides are multilinear in the z_u ; compare coefficients of $\prod_{u \in S} z_u$. $\square$

Remark 2.2. The identity is elementary and product-graph Matrix-Tree computations of this kind appear in Martin–Reiner [3] and Bernardi [1, §4]; we make no novelty claim for Theorem 2.1 itself. Its role here is architectural: it is the exact statement whose bijective realization would prove (1) bijectively (Corollary 2.3 and Problem 5.1). We first found it as an experimental identity: it was confirmed class-by-class for Q_2, Q_3, Q_4 (3+15+255 supports; the Q_4 case over all 679 477 248 tree–root pairs) before the proof was written down.

Corollary 2.3. *Iterating Theorem 2.1 yields (1).*

Proof. Summing the theorem over all S (the $S = \emptyset$ term vanishes) and using $\sum_R z^{|R|} F_G(R) = \det(zI + L_G)$ at $z = 2$:

$$c(G \times K_2) = \frac{1}{2} c(G) \det(2I + L_G).$$

For $G = Q_{n-1}$, whose Laplacian eigenvalues are $2k$ with multiplicity $\binom{n-1}{k}$, $\det(2I + L) = \prod_{k=0}^{n-1} (2k+2)^{\binom{n-1}{k}}$. Writing $R_n = 2^n c(Q_n)$, the recursion reads $R_n = R_{n-1} \prod_{j=1}^n (2j)^{\binom{n-1}{j-1}}$, and Pascal’s rule gives $R_n = \prod_{j=1}^n (2j)^{\binom{n}{j}}$ by induction, which is (1). $\square$

Of course, evaluating $\det(2I + L_{Q_{n-1}})$ spectrally re-uses linear algebra; combinatorially, $\det(2I + L_G) = \sum_S 2^{|S|} F_G(S)$ counts *pairs* (an S -rooted spanning forest, a $\{0, 1\}$ -coloring of its roots), so the recursion has the set-level form

$$(2) \quad \begin{aligned} & \{\text{spanning trees of } G \times K_2\} \times \{0, 1\} \\ \longleftrightarrow & \{\text{spanning trees of } G\} \times \{\text{root-2-colored rooted spanning forests of } G\}, \end{aligned}$$

and a bijective proof of (1) needs exactly a constructive form of (2), level by level.

3. BIJECTIVE FRAGMENTS

3.1. The full-support slice.

Proposition 3.1. *A spanning tree of $G \times K_2$ containing every vertical edge is the same data as a spanning tree of G together with a level in $\{0, 1\}$ for each of its $M - 1$ edges: contract the vertical edges to project, and lift each base tree edge to its recorded level to invert. Hence $N_G(V(G)) = c(G) 2^{M-1}$, the $S = V(G)$ case of Theorem 2.1.*

3.2. Partial doubles.

Definition 3.2. For $S \subseteq V(G)$, the *partial double* $G^{(S)}$ is the graph obtained from G by splitting each $u \notin S$ into u_0, u_1 : an edge (u, v) of G becomes two parallel edges when $u, v \in S$; the two edges $(u, v_0), (u, v_1)$ when $u \in S, v \notin S$; and $(u_0, v_0), (u_1, v_1)$ when $u, v \notin S$.

Contracting the vertical edges of a tree with support S is a bijection onto the spanning trees of $G^{(S)}$, so Theorem 2.1 is equivalent to

$$(3) \quad c(G^{(S)}) = \frac{1}{2} c(G) 2^{|S|} F_G(S).$$

For $S = V(G)$ this is Proposition 3.1 (all edges doubled); for $S = \emptyset$ both sides vanish on connectivity grounds. Splitting one vertex at a time interpolates between $G^{(V)}$ and $G^{(S)}$ and is a natural inductive route to a bijection, though the ratio $F_G(S)/F_G(S \cup u)$ is not local, so the induction must carry more structure than a single count.

3.3. What the spins do and do not give. Bernardi's independence theorem [1, Thm. 1] concerns uniformly random rooted forests conditioned on their projection; transported to spanning trees it says the *spins* (for each $u \in S$, the level of the endpoint of the vertical edge at u nearer the root) should be free bits. We verified computationally: for the ensemble of (tree, root) pairs of Q_3 , conditioned on the direction and the support S , the joint distribution of the spins is *exactly* uniform on $\{0, 1\}^S$ (all 45 conditioned classes, zero discrepancy), and remains consistent with uniform on samples of 2×10^6 (Q_4) and 10^6 (Q_5) uniform trees. For trees with a *fixed* root the uniformity provably fails (e.g. $\chi^2 = 330.7$ on 65 dof already for Q_3). Thus the factor $2^{|S|}$ of Theorem 2.1 is realized by free bits only in the rooted-anywhere ensemble; any bijection must fix the root only after the bits are extracted. This matches the shape of (2), which is stated for a 2-fold cover of the tree set rather than for rooted trees.

4. FINE STRUCTURE OF THE REMAINING STEP

Deleting (rather than contracting) the vertical edges of a tree T with support S leaves horizontal forests H_0, H_1 in the two levels; projecting both to G gives an edge multiset U with multiplicities in $\{1, 2\}$, where a multiplicity-1 edge remembers a free level choice and a multiplicity-2 edge uses both levels. Vertex and edge counts show that T is a tree if and only if the chosen level-lift of (S, U) is connected. Exhaustively on Q_3 (base Q_2):

- every occurring U decomposes as (spanning tree of the base) $\uplus$ (S -rooted spanning forest of the base) — the edge budgets match Theorem 2.1 exactly — and conversely every decomposable (S, U) occurs;
- with m the number of multiplicity-1 edges and d the number of decompositions, every class satisfies the *mass formula* $N(S, U) \cdot d = 2^m$ (the classes split as $d = 1$ with all 2^m lifts connected, and $d = 2$ with exactly half).

On Q_4 (base Q_3 ; all 42 467 328 trees, 2 459 160 classes) the mass formula holds in 95.7% of classes; but the full data reveals a *universal law* subsuming it. In every class of Q_3 and of Q_4 without exception,

$$(4) \quad N(S, U) = d(S, U) \cdot 2^{m-2s} \quad \text{for some integer } s = s(S, U) \geq 0$$

(observed $s \leq 5$ on Q_4): the connected level-lifts distribute over the d decompositions of U in equal shares whose codimension in $\mathbb{F}_2^m$ is always *even*. The mass formula is the special case $d = 2^s$. Probing the geometry: in a sample of 40 exceptional classes, the set $V \subseteq \mathbb{F}_2^m$ of connected lifts was, in 22 cases, exactly a union of d cosets of its stabilizer subspace $W = \{x : V \oplus x = V\}$ with $\dim W = m - 2s$; in the remaining 18, V is a union of fewer, larger cosets with the decompositions clustering in groups of 4 per coset. We conjecture (4) for all connected base graphs, and identifying the invariant $s(S, U)$ together with a canonical assignment of lifts to decompositions is, in our view, the combinatorial heart of (2).

5. OPEN PROBLEMS

Problem 5.1. *Construct an explicit bijection realizing (2) — equivalently (3) for all S — for $G = Q_{n-1}$, or for arbitrary connected G . By Corollary 2.3 this yields a bijective proof of (1), resolving the bijective form of the question of [2, Example 5.6.10].*

Problem 5.2. *Prove the universal law (4) and identify the invariant $s(S, U)$; determine when the connected lifts form exactly d cosets of a subspace of dimension $m - 2s$, and construct the canonical assignment of lifts to decompositions in general.*

Problem 5.3. *Make Bernardi’s spin-flip independence constructive on rooted-anywhere trees (his proof is an induction through vertex deletions and arc mergings), and determine its interaction with the contraction map of Proposition 3.1.*

REFERENCES

- [1] O. Bernardi, *On the spanning trees of the hypercube and other products of graphs*, Electron. J. Combin. 19(4) (2012), #P51.
- [2] R. P. Stanley, *Enumerative Combinatorics, Volume 2*, Cambridge Univ. Press, 1999; 2nd ed. 2024.
- [3] J. L. Martin, V. Reiner, *Factorization of some weighted spanning tree enumerators*, J. Combin. Theory Ser. A 104 (2003) 287–300.